**Department of Electrical and Electronics Engineering**
**Academic Year 2022 – 2023 (Odd Semester)**

**Degree, Semester & Branch: V Semester B.E. EEE**
**Course Code & Title: EE8501 Power System Analysis**
**Name of the Faculty member: Mrs G Sivapriya, AP/EEE**

# Innovative Practice Description

- **Unit / Topic:** Unit I / Y-bus formation-Singular transformation methods
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen: Mind-map**
- **Justification:**
    - A mind map's radial structure directly corresponds to the way our brains store and retrieve information.
    - A mind map conveys hierarchy and relationships between individual ideas, enabling you to see the big picture.
    - After teaching the concept, I thought of conducting this activity for making the students to give the their individual ideas about Y bus formation using singular transformation method. which enhance the learning level and as a teacher I can judge the understanding level of the students.

**Time Allotted for the Activity:** 5 Minutes

After teaching the concept, the students were made to pair with their neighbors

Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about Y-bus formation-Singular transformation methods concept for 2 minute.
- ✓ Then I told them to Pair with their neighbors and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design mind-map and submit within 3 minutes.

- **CO – PO / PSO mapping:**

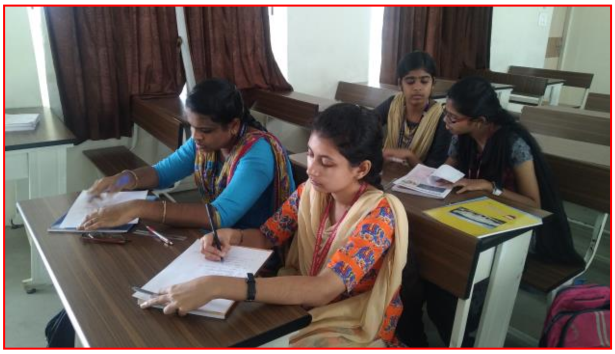
| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO7 | PO12 | PSO1 |
|---|---|---|---|---|---|---|---|---|

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **C301.3** | 3 | 2 | 1 | 1 | 1 | 1 | **1** | 3 |

- **PO / PSO mapped:**

| **Innovative practice** | **PO1** | **PO2** | **PO3** | **PO4** | **PO5** | **PO7** | **PO12** | **PSO1** |
|---|---|---|---|---|---|---|---|---|
| | **3** | **2** | **1** | **1** | **1** | **1** | **1** | **3** |
| **Justification for correlation** | Students will be applying the fundamental knowledge of mathematics, electrical engineering to solve symmetrical fault analysis. Thus PO1 is mapped at level 3. | Students will be analyzing the symmetrical fault current to select the suitable rating of protective systems. Thus, it is mapped to Level23 | Students will be developing the solution for symmetrical fault analysis. Thus, it is mapped to Level 1 | Students will be analyzing the performance of power system briefly by means of interpreting the result of symmetrical fault analysis. Thus it is mapped to Level 1 | Students will be applying thevenin's therorem and bus impedance matrix techniques to solve symmetrical fault analysis using modern tools. Thus it is mapped to Level 1.. | Students need to understand and analyze the challenges of renewable green sources for societal benefits. Thus, it is mapped to Level 1. . | Students will be applying the concept of symmetrical fault analysis in their life and update their knowledge based on recent trends. Thus, it is mapped to Level 1. | Students will be analyzing the performance of electric power system using Symmetrical fault analysis. Thus, it mapped to Level 3. |

- **Images / Screenshot of the practice:**

❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**

- **Benefits:**
  - I preferred this Activity because A mind map makes use of mental triggers (such as pictures, colors and connections) to help your brain memorize things more easily.
  - The best part — mind mapping doesn't feel like work

**Challenges:**

❖ In the class mostly boys hesitate to answer to the questions.

❖ Time utilization for conducting activity.

**Steps taken:**

❖ The boys are sitting in 2 columns – I planned to choose more pairs from boys to involve them in the activity.

2. **Post-implementation Reflection :**

- **Benefits:**

➢ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- **Challenges:**

➢ Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

✓ The assessment of effectiveness of the activity was felt when told most of the points.

✓ While conducting the activity, I understood that the students will be able to explain the concepts of Y-bus formation-Singular transformation methods.

✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 70% of students answered**.

References:

1. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-mini-maps

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# Innovative Practice Description

- **Unit / Topic:** Unit II / Flow chart and Problems in NR method
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 6
- **Activity Chosen: Concept -map**
- **Justification:**
  - Differentiate NR Method and Power flow Analysis
  - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the two concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.

**Time Allotted for the Activity:** 5 Minutes

After teaching the concept, the students were made to pair with their neighbors

Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about Flow chart and Problems in NR method concept for 2 minute.
- ✓ Then I told them to Pair with their neighbors and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design Concept-map and submit within 3 minutes.

- **CO – PO / PSO mapping:**

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO7 | PO12 | PSO1 |
|---|---|---|---|---|---|---|---|---|
| **C301.3** | 3 | 1 | 1 | 2 | 1 | 1 | **1** | 3 |

- **PO / PSO mapped:**

| Innovative practice | PO1 | PO2 | PO3 | PO4 | PO5 | PO7 | PO12 | PSO1 |
|---|---|---|---|---|---|---|---|---|
| | **3** | **3** | **1** | **2** | **1** | **1** | **1** | **2** |
| **Justification for correlation** | Students will be applying the fundamental knowledge of mathematics, electrical engineering to solve symmetrical fault analysis. Thus PO1 is mapped at level 3. | Students will be analyzing the symmetrical fault current to select the suitable rating of protective systems. Thus, it is mapped to Level 3 | Students will be developing the solution for symmetrical fault analysis. Thus, it is mapped to Level 1 | Students will be analyzing the performance of power system briefly by means of interpreting the result of symmetrical fault analysis. Thus it is mapped to Level 1 | Students will be applying thevenin's therorem and bus impedance matrix techniques to solve symmetrical fault analysis using modern tools. Thus it is mapped to Level 1.. | Students need to understand and analyze the challenges of renewable green sources for societal benefits. Thus, it is mapped to Level 1. . | Students will be applying the concept of symmetrical fault analysis in their life and update their knowledge based on recent trends. Thus, it is mapped to Level 1. | Students will be analyzing the performance of electric power system using Symmetrical fault analysis. Thus, it mapped to Level 3. |

- **Images / Screenshot of the practice:**

❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**

- **Benefits:**
- I preferred this Activity because Concept mapping helps to elicit my students' thinking and relationships between concepts and ideas. In the most successful cases, concept mapping when used as a strategy with groups, the tasks involved helps to generate deep and meaningful conversations among students.
- When used for a specific topic like comparing two concepts, concept mapping works well when there are some constraints on the nodes and the relational links. Specifically, try to limit the number of concepts (nodes) and ask students to elaborate on their relational links.

- **Challenges:**

❖ While it is a strategy that can be used for individual learning, its benefits are most effective for collaborative work but this needs to be well defined. Specifically, concept mapping can become a trivial task if concepts are not well defined. And, when done in groups, students need to be encouraged to engage in such a way that no one person takes over. Both of these are challenging.

❖ In the class mostly boys hastate to answer to the questions.

❖ Time utilization for conducting activity.

**Steps taken:**

❖ The boys are sitting in 2 columns – I planned to choose more pairs from boys to involve them in the activity.

2. **Post-implementation Reflection :**

- **Benefits:**

➢ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- **Challenges:**

➢ Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

✓ The assessment of effectiveness of the activity was felt when told most of the points.

✓ While conducting the activity, I understood that the students will be able to explain the concepts of Watch Dog Timer and Real Time Clock.

✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 50% of students answered**.

References:

1.https://www.saltise.ca/teaching-resources/strategies/concept-mapping/

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**Name of the Faculty member (s): Mrs.G.Sivapriya**

**Innovative Practice Description**

- **Unit / Topic:** Unit III / Symmetrical Fault Analysis- Thevenin's Theorem
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen:** Power System Simulator Tool
- **Justification:**

Symmetrical Fault analysis is the method of finding symmetrical fault current using different methods such as thevenin's theorem, Z bus algorithm and KCL, KVL.IN this activity, short circuit current or symmetrical fault current has been found by using Power system simulator tool. By learning this student will able to decide how the fault current increase when fault occurs.

- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
    - Symmetrical Fault problems were explained and the students were taken to PSS Lab.
    - A sample simulation were simulated in the lab and their doubts were clarified
    - A Problem was given to the students and asked to simulate individually and their simulation outputs were individually verified
    - Students submitted their work as a report and it is evaluated.

- **CO – PO / PSO mapping:**

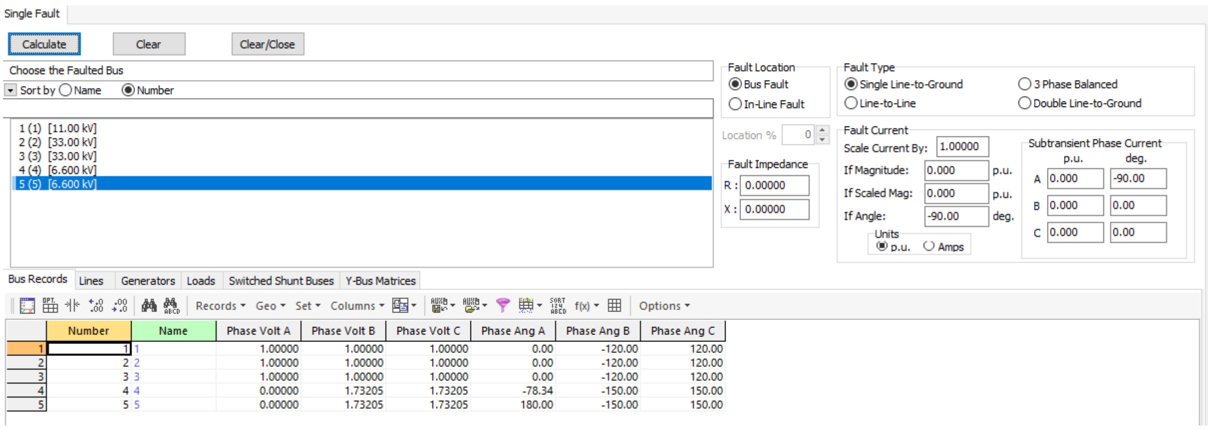
| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO7 | PO12 | PSO1 |
|---|---|---|---|---|---|---|---|---|
| C301.3 | 3 | 3 | 1 | 2 | 1 | 1 | 1 | 3 |

- **PO / PSO mapped:**

| Innovative practice | PO1 | PO2 | PO3 | PO4 | PO5 | PO7 | PO12 | PSO1 |
|---|---|---|---|---|---|---|---|---|
| | 3 | 3 | 1 | 2 | 1 | 1 | 1 | 2 |
| Justification for correlation | Students will be applying the fundamental knowledge | Students will be analyzing the symmetrical | Students will be developing the solution | Students will be analyzing the performance | Students will be applying thevenin's therorem | Students need to understand and | Students will be applying the concept of | Students will be analyzing the performance |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | of mathematics, electrical engineering to solve symmetrical fault analysis. Thus PO1 is mapped at level 3. | fault current to select the suitable rating of protective systems. Thus, it is mapped to Level 3 | for symmetrical fault analysis. Thus, it is mapped to Level 1 | of power system briefly by means of interpreting the result of symmetrical fault analysis. Thus it is mapped to Level 1 | and bus impedance matrix techniques to solve symmetrical fault analysis using modern tools. Thus it is mapped to Level 1.. | analyze the challenges of renewable green sources for societal benefits. Thus, it is mapped to Level 1. . | symmetrical fault analysis in their life and update their knowledge based on recent trends. Thus, it is mapped to Level 1. | of electric power system using Symmetrical fault analysis. Thus, it mapped to Level 3. |

- **Images / Screenshot of the practice:**




- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - While solving Student felt that their computational time was reduced due to simulation and they were able to solve many problems quickly with the help of simulation.
  - ❖ ***Benefit of the practice:***
    - Student understood the Symmetrical Fault Analysis problem clearly and able to explain the procedure of calculation whenever asked.
    - Most of the students have attended the question in their Internal Assessment Test.
    - Student Interest towards simulation tool is increased
  - ❖ ***Challenges faced in implementation:***

- Initially Few students took more time for their simulation and later by practice their time for completing the simulation is reduced

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**Innovative Practice Description**

- **Unit / Topic:** Unit IV / Un Symmetrical Fault Analysis
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** ETAP Simulation Tool
- **Justification:**

Unsymmetrical Fault analysis is the method of finding Unsymmetrical fault current using Symmetrical Components. There are different types of unsymmetrical faults such as L-G, L-L, LL-G faults can be found by using ETAP simulator tool. By learning this student will able to decide how the fault current increase when fault occurs.

- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
    - Unsymmetrical Fault problems were explained and the students were taken to PSS Lab.
    - A sample simulation were simulated in the lab and their doubts were clarified
    - A Problem was given to the students and asked to simulate individually and their simulation outputs were individually verified
    - Students submitted their work as a report and it is evaluated.

- **CO – PO / PSO mapping:**

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO7 | PO12 | PSO1 |
|---|---|---|---|---|---|---|---|---|
| C301.3 | 3 | 3 | 1 | 2 | 1 | 1 | **1** | 3 |

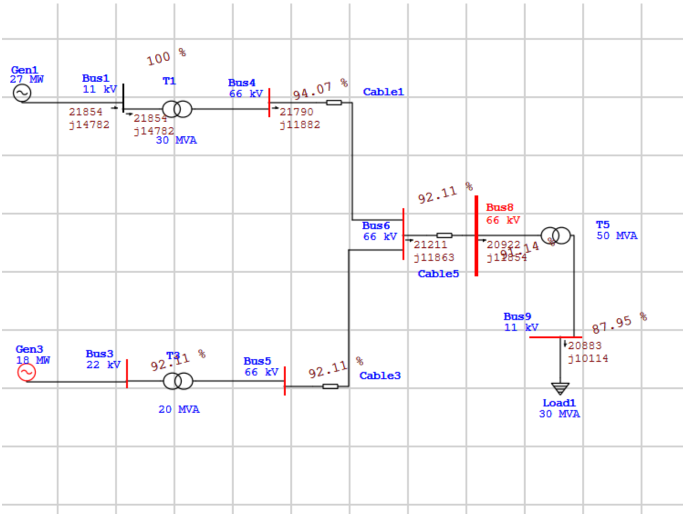
- **PO / PSO mapped:**

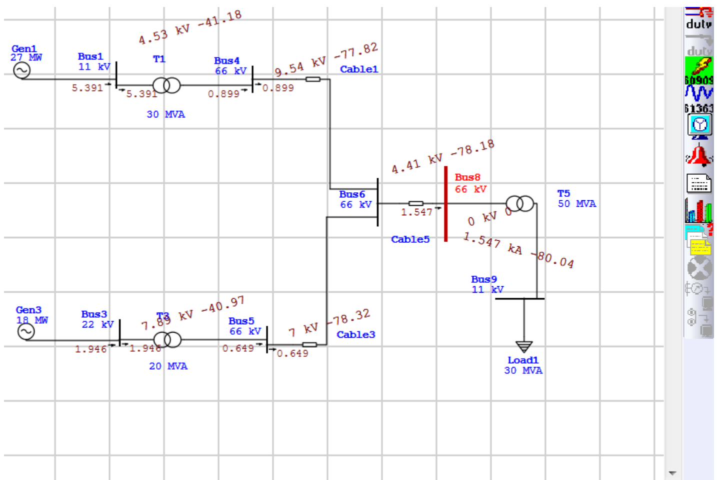
| Innovative practice | PO1 | PO2 | PO3 | PO4 | PO5 | PO7 | PO12 | PSO1 |
|---|---|---|---|---|---|---|---|---|
| | 3 | 3 | 1 | 2 | 1 | 1 | 1 | 2 |
| Justification for correlation | Students will be applying the fundamental knowledge of | Students will be analyzing the symmetrical fault current | Students will be developing the solution for | Students will be analyzing the performance of power | Students will be applying thevenin's therorem and bus | Students need to understand and analyze | Students will be applying the concept of symmetrical | Students will be analyzing the performance of electric |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | mathematics, electrical engineering to solve symmetrical fault analysis. Thus PO1 is mapped at level 3. | to select the suitable rating of protective systems. Thus, it is mapped to Level 3 | symmetrical fault analysis. Thus, it is mapped to Level 1 | system briefly by means of interpreting the result of symmetrical fault analysis. Thus it is mapped to Level 1 | impedance matrix techniques to solve symmetrical fault analysis using modern tools. Thus it is mapped to Level 1.. | the challenges of renewable green sources for societal benefits. Thus, it is mapped to Level 1. . | fault analysis in their life and update their knowledge based on recent trends. Thus, it is mapped to Level 1. | power system using Symmetrical fault analysis. Thus, it mapped to Level 3. |

- **Images / Screenshot of the practice:**

**Before Fault** **After Fault**




- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - While solving Student felt that their computational time was reduced due to simulation and they were able to solve many problems quickly with the help of simulation.
  - ❖ ***Benefit of the practice:***
    - Student understood the Unsymmetrical Fault Analysis problem clearly and able to explain the procedure of calculation whenever asked.

- Most of the students have attended the question in their Internal Assessment Test.
- Student Interest towards simulation tool is increased

❖ ***Challenges faced in implementation:***

- Initially Few students took more time for their simulation and later by practice their time for completing the simulation is reduced

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## Innovative Practice Description

- **Unit / Topic:** Unit V / Solution of Swing Equation using Step by Step Method
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen: Think-Pair-Share**
- **Justification:**
  - Differentiate between two different methods of solutions obtained from swing equations such as step by step method and Modified Eulers method.
  - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the different methods which enhance the learning level and as a teacher I can judge the understanding level of the students.

**Time Allotted for the Activity:** 6 Minutes
After teaching the concept, the students were made to pair with their neighbours.
Total Strength is 29, Number of Pairs – 14
Photographer: one student – Mr . M. Vignesh Kumar (interested in photography)
Reporter: Myself

At the end the Class (Last 6 minutes)
- ✓ I asked the students to think about step by step method and Modified Eulers method.for 2 minute.
- ✓ Then I told them to Pair with their neighbours and discuss about the difference between the models for another 1 minute.
- ✓ Finally, I selected on Pair from each column randomly and ask them to share one difference between the concepts. (3 miuntes)
- ✓ When each pair told the points simultaneously, I wrote the points on the board.
- ✓ Finally, I summarized the points and I told one missed point to all the students.
- ✓ With the help of PPT prepared (first slide- Activity name, Second slide-Topic for Think-pair-share, Third slide - answer) the answers were be projected to the students.
- ✓ After completion of the class I uploaded the PPT in the course website (Canvas)

- **CO – PO / PSO mapping:**

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO7 | PO12 | PSO1 |
|---|---|---|---|---|---|---|---|---|
| **C301.3** | 3 | 3 | 1 | 1 | 1 | 1 | **1** | 3 |

- **PO / PSO mapped:**

| **Innovative practice** | **PO1** | **PO2** | **PO3** | **PO4** | **PO5** | **PO7** | **PO12** | **PSO1** |
|---|---|---|---|---|---|---|---|---|
| | **3** | **3** | **1** | **1** | **1** | **1** | **1** | **2** |
| **Justification for correlation** | Students will be applying the fundamental knowledge of mathematics, electrical engineering to solve symmetrical fault analysis. Thus PO1 is mapped at level 3. | Students will be analyzing the symmetrical fault current to select the suitable rating of protective systems. Thus, it is mapped to Level 3 | Students will be developing the solution for symmetrical fault analysis. Thus, it is mapped to Level 1 | Students will be analyzing the performance of power system briefly by means of interpreting the result of symmetrical fault analysis. Thus it is mapped to Level 1 | Students will be applying thevenin's therorem and bus impedance matrix techniques to solve symmetrical fault analysis using modern tools. Thus it is mapped to Level 1.. | Students need to understand and analyze the challenges of renewable green sources for societal benefits. Thus, it is mapped to Level 1. . | Students will be applying the concept of symmetrical fault analysis in their life and update their knowledge based on recent trends. Thus, it is mapped to Level 1. | Students will be analyzing the performance of electric power system using Symmetrical fault analysis. Thus, it mapped to Level 3. |

- **Images / Screenshot of the practice:**

❖ *Reflective Critique:*

**1. Pre-implementation Reflection :**

I preferred this Activity because they have step by step method and Modified Eulers method.separately.

**Challenges anticipated:**

❖ In the class mostly boys hastate to answer to the questions.

❖ Time utilization for conducting activity.

**2. Post implementation Reflection:**

- **Benefits:**

➢ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- **Challenges:**

➢ Slow learners were not able to understand some topics during discussion hours.

*Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

✓ The assessment of effectiveness of the activity was felt when told most of the points.

✓ While conducting the activity, I understood that one point was missed with respect to method of computation and finally I insisted the point they missed to tell.

✓ The success of the activity was evaluated by asking the same question in Internal

✓ Assessment test III – Around 30% of students answered correctly.

**References:**

1. http://www.readwritethink.org/professional-development/strategy-guides/using-think- pair-share-30626.html
2. https://in.video.search.yahoo.com – Think pair share

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